ALTAF SHAIKH

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Summary

Final Year B.Tech Computer Science & Engineering student and **Software Engineer** with hands-on experience designing, developing, and deploying full-stack applications using JavaScript, TypeScript, React.js, Node.js, Express.js, MongoDB, and Python. Experienced in building **REST APIs, real-time collaborative systems, authentication workflows, database-driven applications, and AI-powered software features**. Strong foundation in **Data Structures & Algorithms, Object-Oriented Programming, DBMS, and Operating Systems**, with experience solving engineering problems, debugging applications, and working with Git-based development workflows. Smart India Hackathon (SIH) 2025 Finalist and Google Student Ambassador with a strong interest in building reliable, scalable, and user-focused software products.

Technical Skills

Frontend Development: React.js, JavaScript, TypeScript, HTML5, CSS3, Responsive Web Design, Tailwind CSS, Bootstrap
Backend & APIs: Node.js, Express.js, REST APIs, JWT Authentication
Databases: MongoDB
Programming Languages: JavaScript, TypeScript, Python, Java
AI & Machine Learning: LangChain, OpenAI API, Hugging Face, Sentence Transformers, FAISS, Ollama, NumPy, Pandas, Scikit-learn (Basics)
Real-Time & Collaboration: Socket.IO, Yjs, WebRTC
Tools & Design: Git, GitHub
Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

Projects

• **SatQuery AI — Agentic Vision-Language Assistant** *GitHub*
Python, LangChain, OpenAI-compatible APIs, Hugging Face, FAISS, Sentence Transformers, Ollama, pytest, Ruff

- Built a research-oriented **agentic AI framework** that connects natural-language remote-sensing queries with retrieval, reasoning, and extensible specialist-tool workflows through an **agent controller**.
- Implemented modular **agent orchestration, knowledge-space and solution-space retrieval, toolkit registry, and configurable LLM integration** for extensible remote-sensing workflows.
- Developed retrieval components using **Sentence Transformers, embeddings, FAISS vector search, and DualRAG** for knowledge and solution retrieval.
- Added configurable support for **OpenAI APIs, Hugging Face, LangChain, and Ollama**, with automated testing through **pytest** and code-quality checks using **Ruff**.

• **SyncSpace — AI-Powered Collaborative Development Platform** *GitHub* | *Live Demo*
React, Vite, Node.js, Express.js, MongoDB, Socket.IO, Yjs, WebRTC, OpenAI API

- Architected a full-stack collaborative workspace combining **real-time coding, team communication, video meetings, whiteboarding, file sharing, and AI-assisted development**.
- Built a modular Node.js/Express backend with versioned **REST APIs**, authentication, room management, document handling, file management, code execution, replay, and analytics services.
- Implemented **AI Code Reviewer** and **AI Session Summary** workflows using OpenAI integration with local fallback behavior for supported functionality.
- Integrated **Socket.IO, Yjs, WebRTC, React Konva, and MongoDB** for real-time collaboration, synchronized editing, peer-to-peer communication, session replay, and analytics.

• **ApnaVault — Personal Document Management Platform** *GitHub* | *Live Demo*
React, TypeScript, Node.js, Express.js, MongoDB Atlas, Supabase Storage

- Built a secure document management platform that helps individuals and families organize, store, and manage important documents with folder management and a centralized dashboard.
- Implemented document upload and management using React, TypeScript, Node.js, Express.js, MongoDB Atlas, and private Supabase Storage, with authentication and workspace-based access controls.
- Developed dashboard statistics, family member management, document expiry reminders, notifications, and storage usage tracking to simplify document organization and management.

Education

Sanjeevan Group of Institutions **2023 – 2027 (Expected)**
Bachelor of Technology (B.Tech) in Computer Science and Engineering *India*

- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

Achievements

- **Smart India Hackathon (SIH) 2025** – National-Level Finalist
- **BIG-O Battle: Vibe Coding SaaS Startup Challenge**, GDG VIT Mumbai – Top 2 Position
- **Buildverse Hackathon 2025**, HackWithIndia and GeeksforGeeks – Top 5 Position
- Participated in Google Cloud Agentic AI Day, powered by Hack2skill

Leadership and Activities

- **Google Student Ambassador (Team Gemini)**, Google – May 2026 – Present
- **Event Coordinator (Tech Management – Parikrama)**, Sanjeevan Group of Institutions – March 2026